Machine Learning Assignment :- 3

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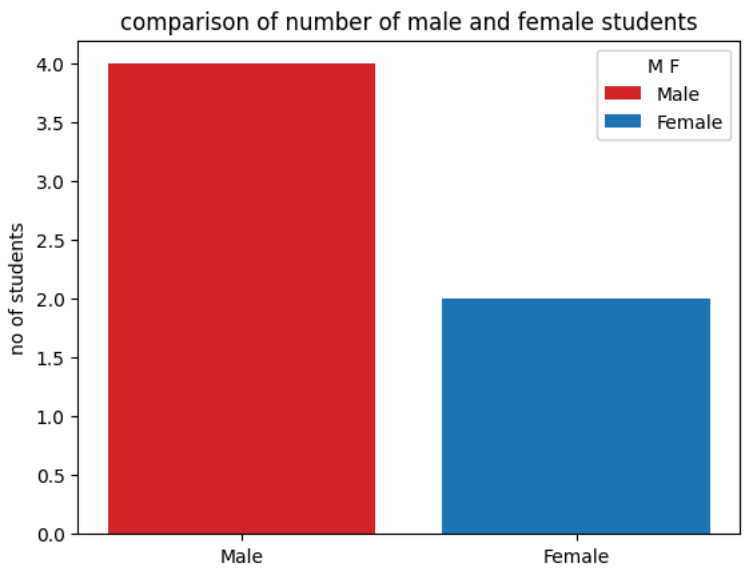
Visualize the data using Python by plotting the graphs for assignment no. 1 and 2. Consider a suitable data set. Use Scatter plot, Bar plot, Box plot, Pie chart, Line Chart.

```
from numpy import nan
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
data={
    'Name':["Harish","Ram","shital","Riya","OM","Rohan"],
    'Age':[18,19,20,21,18,18],
    'Gender':['M','M','F','F','M','M'],
    'Marks':[90,80,50,60,10,75],
    'Internship':[1.0,0.0,1.0,1.0,0.0,0.0]
}
df_f=pd.DataFrame(data)
male_count=np.sum(df_f['Gender']=='M')
female_count=np.sum(df_f['Gender']=='F')
```

Bar Graph (Number of boys and girls)

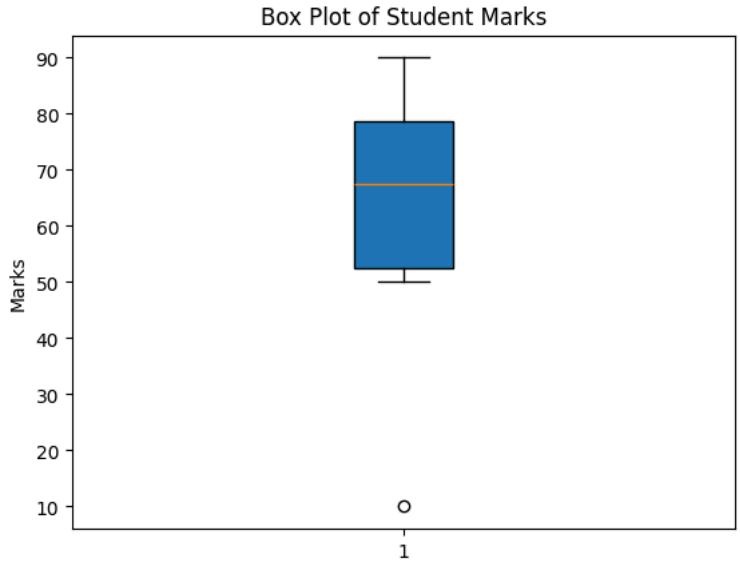
```
fig, ax = plt.subplots()

counts=[male_count,female_count]
bar_labels=['Male','Female']
bar_colors=['tab:red','tab:blue']
# ax.bar(df_f.Gender,counts,label=bar_labels,color=bar_colors)
ax.bar(bar_labels,counts,label=bar_labels,color=bar_colors)
ax.set_ylabel('no of students')
ax.set_title('comparison of number of male and female students')
ax.legend(title='M F')
plt.show()
```



BOX Plot (For marks of student)

student with 10 marks is an outlier

```
marks=df_f['Marks']
plt.boxplot(marks,vert=True,patch_artist=True,label=['marks'])
plt.title('Box Plot of Student Marks')
plt.ylabel('Marks')
plt.show()
```



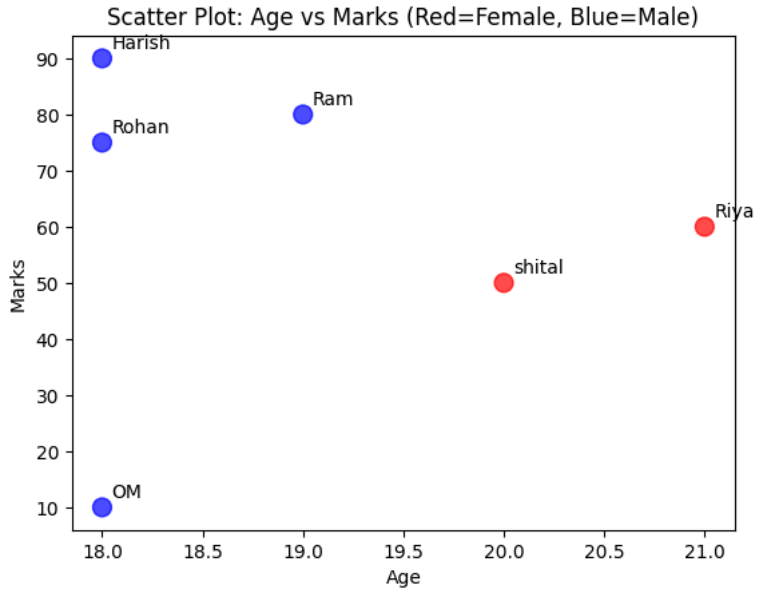

```
stats = {
    'min': np.min(marks),
    'q1': np.percentile(marks, 25),
    'median': np.median(marks),
    'q3': np.percentile(marks, 75),
    'max': np.max(marks)
}
print(stats)
```

```
{'min': 10, 'q1': np.float64(52.5), 'median': np.float64(67.5), 'q3': np.float64(78.75), 'max': 90}
```

scatter plot

```
df=df_f


colors = df['Gender'].map({'M': 'blue', 'F': 'red'})
plt.scatter(df['Age'], df['Marks'], c=colors, s=100, alpha=0.7)

# Add Student Names as labels
for i, txt in enumerate(df['Name']):
    plt.annotate(txt, (df['Age'][i], df['Marks'][i]), xytext=(5,5), textcoords='offset points')

plt.title('Scatter Plot: Age vs Marks (Red=Female, Blue=Male)')
plt.xlabel('Age')
plt.ylabel('Marks')
```

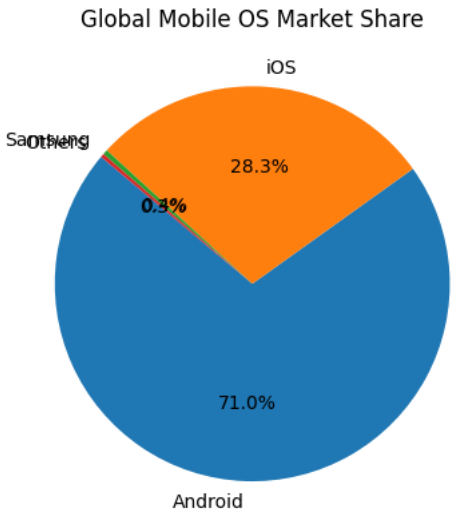
```
Text(0, 0.5, 'Marks')
```



Pie chart

display os market share of different operating systems

```
import pandas as pd
import matplotlib.pyplot as plt

data_pie = {
    'OS': ['Android', 'iOS', 'Samsung', 'Others'],
    'Market_Share': [71.0, 28.3, 0.4, 0.3]
}
df_pie = pd.DataFrame(data_pie)

plt.pie(df_pie['Market_Share'], labels=df_pie['OS'], autopct='%1.1f%%', startangle=140)
plt.title('Global Mobile OS Market Share')
plt.show()
```



Line chart

(Display How revenue changes monthly)

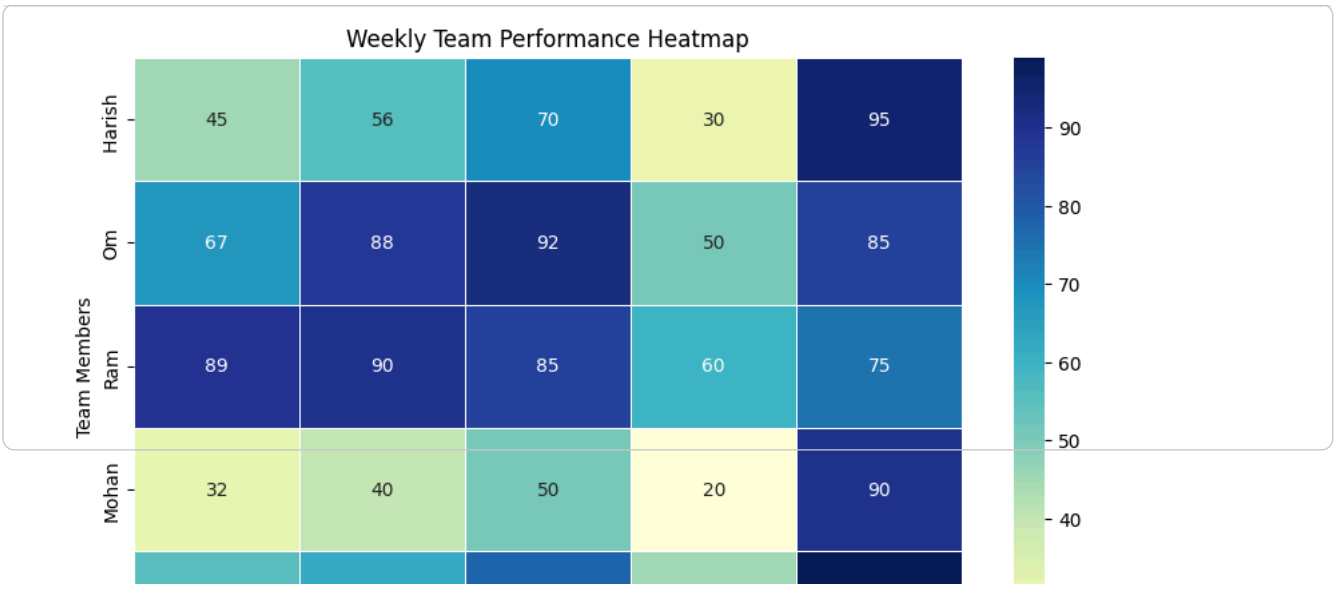
```
import pandas as pd
import matplotlib.pyplot as plt

data_line = {
    'Month': ['Jan', 'Feb', 'Mar', 'Apr', 'May', 'Jun'],
    'Revenue_USD': [12000, 15000, 14500, 19000, 24000, 28000]
```

```
}
df_line = pd.DataFrame(data_line)

plt.plot(df_line['Month'], df_line['Revenue_USD'], marker='o', linestyle='-', color='b')
plt.title('Monthly Revenue Growth')
plt.xlabel('Month')
plt.ylabel('Revenue ($)')
plt.grid(True)
plt.show()
```

Heat map

Performance of team members in a week

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt


data = {
    'Monday': [45, 67, 89, 32, 55],
    'Tuesday': [56, 88, 90, 40, 62],
    'Wednesday': [70, 92, 85, 50, 78],
    'Thursday': [30, 50, 60, 20, 45],
    'Friday': [95, 85, 75, 90, 99]
}


df = pd.DataFrame(data, index=['Harish ', 'Om', 'Ram', 'Mohan', 'Sam'])

plt.figure(figsize=(10, 6))


sns.heatmap(df, annot=True, cmap='YlGnBu', linewidths=0.5)


plt.title('Weekly Team Performance Heatmap')
plt.xlabel('Days of the Week')
plt.ylabel('Team Members')

plt.show()
```


Weekly Team Performance Heatmap
Team Members
Harish
Om
Ram
Mohan
45
56
70
30
95
67
88
92
50
85
89
90
85
60
75
32
40
50
20
90
90
80
70
60
50
40